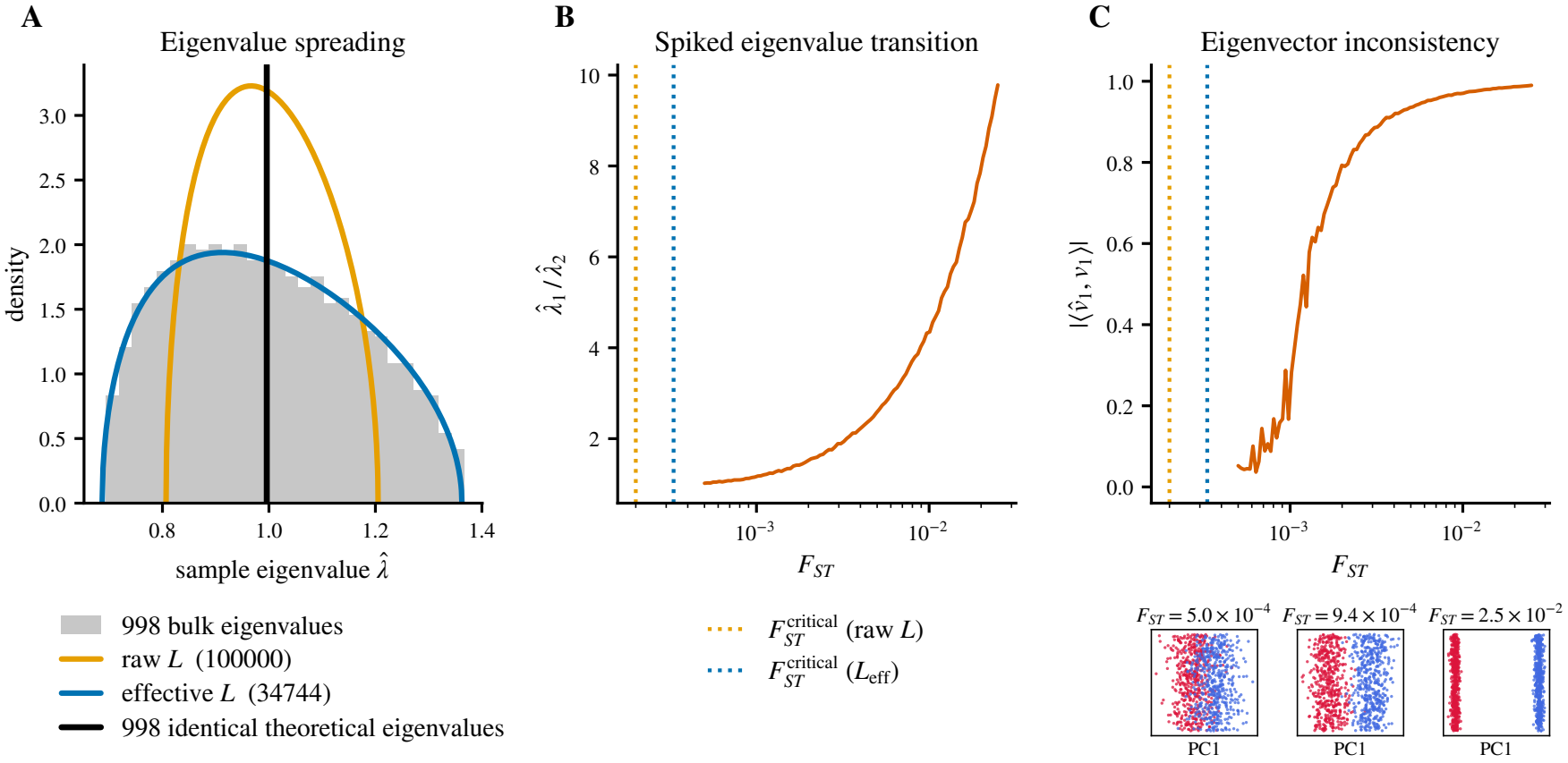

A
Eigenvalue spreading
density
sample eigenvalue $\hat{\lambda}$
998 bulk eigenvalues
raw L (100000)
effective L (34744)
998 identical theoretical eigenvalues
B
Spiked eigenvalue transition
$\hat{\lambda}_1 / \hat{\lambda}_2$
F_{ST}
F_{ST}^{critical} (raw L)
F_{ST}^{critical} (L_{eff})
C
Eigenvector inconsistency
$|\langle \hat{v}_1, v_1 \rangle|$
F_{ST}
$F_{ST} = 5.0 \times 10^{-4}$
$F_{ST} = 9.4 \times 10^{-4}$
$F_{ST} = 2.5 \times 10^{-2}$
PC1
PC1
PC1